

WASP-52b

R-0492

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_250803 · created 2026-07-20T10:29:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460890.7956 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.013
Transit depth (Rp/Rs)^2	2.36 +/- 0.4 [%]
Orbital Inclination (inc)	85.38 +/- 0.99
Airmass coefficient 1 (a1)	1.18 +/- 0.015
Airmass coefficient 2 (a2)	-0.1309 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	None
Optimal Aperture	4.59
Optimal Annulus	16.2
Transit Duration (day)	0.0741 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.013 vs 0.1646 (deviation 0.89σ)
Duration fit vs published	0.0741 d vs 0.0758 d (deviation 0.28σ)
Mid-time O–C	2.3 min
Depth SNR	11.8
Residual scatter	1.25 %

Light curve

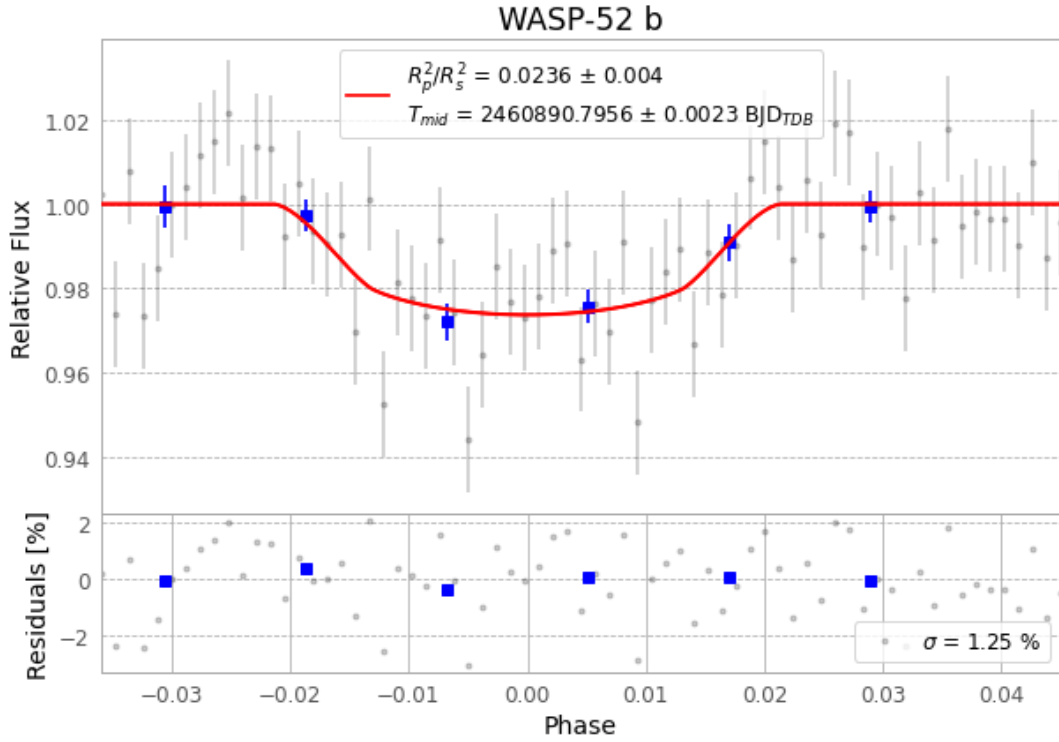


Figure 1 — FinalLightCurve_WASP-52 b_2025-08-02.png (SHA-256 3bf97152280f3ad9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 758bf6092e859249...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), WASP-52250803052715.FITS → WASP-52250803085116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-250803.log (08ece3bf3efd...), FinalLightCurve_WASP-52 b_2025-08-02.png (3bf97152280f...), FinalLightCurve_WASP-52 b_2025-08-02.csv (561f547c7d7b...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 10:29Z.